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Entomological Evaluation of Malaria Vectors at Different Altitudes in Hai District, Northeastern Tanzania

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ABSTRACT Entomological monitoring in four villages situated along an altitude transect in the Hai District of Northeastern Tanzania identified *Anopheles arabiensis* Patton as the principal vector of malaria and detected seasonal changes in vector behavior. Over a 13-mo sampling period, 10,557 mosquitoes were collected with CDC light traps, pyrethrum spray catches, and pit traps of which 5,969 (56.5%) were *An. arabiensis*, 762 (7.2%) were *Anopheles funestus* Giles s.l., 3,578 (33.9%) were culicines, and 248 (2.3%) were nonvector anophelines. Vector densities declined rapidly with increasing altitude, demonstrating a 50% decrease in annual human biting rate for every 86-m rise in altitude. Light traps were found to be more efficient than spray catches for the collection of *An. arabiensis*. This observation was attributed to increased exophily of this species, most notably in the wet season, and is supported by seasonal changes in the human blood index and fed/gravid ratio. These results indicate that spray catches may underestimate the abundance of exophilic vectors such as *An. arabiensis* and that entomological monitoring may require more than one collection method, especially at low vector densities. The annual entomological inoculation rate (EIR) decreased sharply with increasing altitude, with large variation around the estimate at low vector densities. Increased transmission because of unpredictable short rains at low altitudes and spatial clustering of infective mosquitoes may contribute to elevated EIR estimates.

KEY WORDS *Anopheles arabiensis*, malaria, entomological inoculation rate, altitude, Tanzania

The most important vectors of malaria in sub-Saharan Africa belong to the *Anopheles gambiae* Giles complex. In Tanzania, the main malaria vectors are *An. gambiae* Giles, *Anopheles arabiensis* Patton, and *Anopheles funestus* Giles (Mnzava and Kilama 1986), whereas *Anopheles merus* Dönitz may play a role in malaria transmission in the coastal areas (Temu et al. 1998). The proportion of *An. arabiensis* generally increases from the humid coastal areas to the arid inland steppe, whereas the frequency of *An. gambiae* s.s. decreases (Mnzava and Kilama 1986), because of the better adaptation of the former species to arid environments (White 1974, Coetzee et al. 2000). *An. arabiensis* exhibits greater ecological versatility than other members of the *An. gambiae* complex (White 1974) and demonstrates variability in feeding and resting behavior (Githeko et al. 1994). The individual species of the *An. gambiae* complex differ in host-feeding preferences, resting behavior, and vectorial capacity (Petrarca et al. 1991), and these variations in behavior have implications for malaria transmission as well as vector control and selection for insecticide resistance. Proper characterization of vector populations should assist in the development of targeted control strategies and provide a better understanding of the environmental factors that contribute to malaria transmission.

Malaria transmission intensity is best measured by the entomological inoculation rate (EIR) (MacDonald 1957), which can be directly estimated in the field and provides an independent parameter to relate to clinical and parasitological malaria indices. EIRs vary widely both spatially and temporally, from zero to 884 infectious bites per person per year (ib/p/yr) in Africa, with rural areas experiencing higher intensities of transmission than urban areas (Hay et al. 2000). The relationship between transmission intensity and disease burden is complex and an apparent paradox of increased malaria burden at lower transmission intensities has been suggested (Snow et al. 1997). Therefore, the control measures used to reduce transmission, such as insecticide-treated nets (ITN), and their efficacy may vary depending on the level of malaria endemicity (Snow et al. 1997, Snow and Marsh 1998), although similar benefits to young children under very different EIRs have been achieved with ITN (Ellman et al. 1998, Maxwell et al. 2003). To further address this issue it is useful to compare ma-

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laria indices in populations where there are wider differences in the intensity of transmission.

This study represents the entomological component of a project assessing malaria morbidity and mortality at different transmission intensities in the Hai District of Tanzania. We have measured EIR and describe vector dynamics in villages at different altitudes, to identify environmental factors that contribute to malaria transmission and provide an index of transmission intensity to relate to disease burden. The variable behavior of *An. arabiensis* is investigated, and implications for malaria control in areas where this species is the principal vector are discussed.

Materials and Methods

Study Area. The Hai District is a rural area situated on the southwestern slope of Mount Kilimanjaro, where the prevalence of malaria varies between different altitude zones. Malaria transmission in the Hai District is classified as moderately high or "meso-hyperendemic" in lower altitude villages (750–1,000 m), moderate or "mesoendemic" in mid-altitude villages (1,001–1,250 m), low moderate or "hypoendemic" in upper altitude villages (1,251–1,500 m), and non-endemic in highland villages (>1,500 m) (Chandramohan et al. 2001). Four villages were selected for entomological evaluation to represent each of the different altitude strata. At the lowest altitude ranges, the villages Rundugai (850–870 m) and Kware (1,024–1,078 m) are located in the savannah and foothills, respectively, whereas the highland villages of Sonu (1,280–1,404 m) and Foo (1,500–1,674 m) are situated in rain forest (Fig. 1).

Meteorological Recordings. Data from national meteorological stations, Kilimanjaro international airport (890 m) for the lower altitude ranges and Lyamungu coffee estate (1,310 m) for highland sites, were used to provide year averages.

House Selection. Five houses in the lower and mid-altitude villages, Rundugai and Kware, and 10 houses in the upper altitude and highland villages, Sonu and Foo, were randomly selected for light trap catches (LTC). In each house, a bedroom that was occupied by one sleeper was selected for mosquito collection and an untreated bed-net given to the occupant. For five of the selected light trap houses in each village, a neighboring house was designated for pyrethrum spray catch (PSC). Three pit traps (PIT) were constructed in each village, each situated in a shaded area 20–30 m from a spray catch house.

Mosquito Collections. Mosquitoes were collected fortnightly with five to 10 light traps, five pyrethrum spray catches, and three pit traps in each village from June 2001 to June 2002. CDC miniature light traps (model 512, John W. Hock Company, Gainesville, FL) were hung from the ceiling at the foot of the bed. Trained staff set the traps the preceding day, and the battery was operated from sunset to sunrise. Mosquitoes were collected each morning, at which time spray

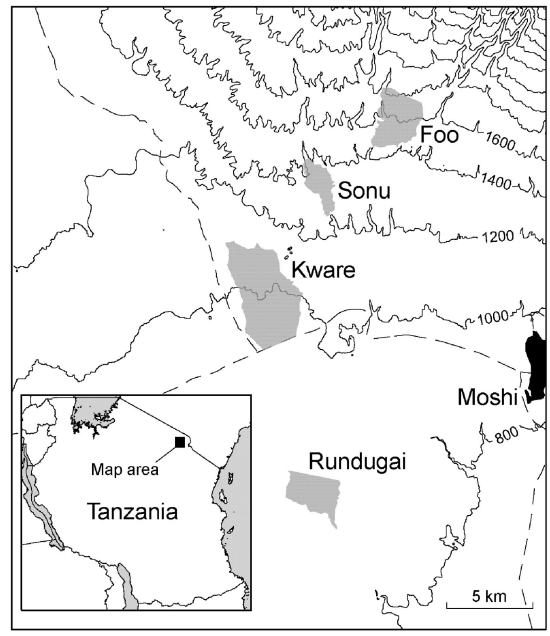


Fig. 1. Map of the Hai District showing the position of four transect villages where entomological sampling was conducted between June 2001 and June 2002. Altitude contours are shown at 200-m intervals; the scale bar represents 5 km. The town of Moshi is shown for reference.

catches also were performed in designated houses. The floor and other surfaces in spray catch rooms were covered with white sheets, and the room was sprayed with pyrethrum (0.4% volume diluted in kerosene). Knocked down mosquitoes were collected after 10 min. The same morning, mosquitoes were collected from pit traps using a 10-min search with a mouth aspirator.

Mosquito Processing. Mosquitoes from all catches were sorted and counted in the laboratory. *An. gambiae* s.l. and *An. funestus* s.l. were identified morphologically (Gillies and Coetzee 1987) and classified by abdominal condition. Four categories were represented among female specimens: unfed, freshly blood-fed, semigravid, and fully gravid. Female mosquitoes were dried on silica gel and stored at 4°C; males, culicines, and nonvector anophelines were discarded after counting. The head and thorax of all malaria vectors caught in light traps and those of the unfed portion of vectors caught in spray catches or pit traps were tested for *Plasmodium falciparum* circumsporozoite protein by enzyme-linked immunosorbent assay (ELISA) (Wirtz et al. 1987). Wells were considered positive when the optical density exceeded the mean +3 standard deviations of a set of three negative controls on the same plate. All positive samples were retested for confirmation. Bloodmeals from freshly blood-fed mosquitoes caught by spray catch or pit trap were smeared onto filter paper and stored at 4°C for later identification of bloodmeal source by precipitin test (Bray et al. 1984). A sample of 442 *An. gambiae* s.l.

was randomly selected for identification of sibling species by using the polymerase chain reaction (PCR) method of Scott et al. (1993). DNA was extracted from a 50-μl aliquot of ELISA sample preparation (head and thorax of individual mosquitoes homogenized in grinding buffer) by using the method described by Collins et al. (1987).

Data Analysis. *Vector Density.* Monthly vector densities in each village were calculated as the geometric means (GM) of the monthly averages of the traps in each village. The human biting rate was derived from GM light trap densities assuming only one sleeper occupied each bedroom, multiplying by a conversion factor of 1.6 to account for the proportional efficiency of light traps relative to human landing catches (Lines et al. 1991). A repeated measures analysis of variance (ANOVA) (SAS Institute 2000) was used to test the effect of time and altitude on the light trap density of each mosquito species. Mean trap densities were $\log_{10}(x + 1)$ transformed and tested for normality before analysis. Seasonal variations in mosquito density from spray catches and pit traps, and seasonal variation in human blood index and sporozoite rate were tested by chi-square analysis.

Fed/Gravid Ratio. The seasonal fed/gravid ratio was determined for indoor- and outdoor-resting vectors from the proportions of vectors collected in the fed and gravid conditions from PSC and PIT, respectively. All abdominal conditions later than freshly blood-fed were included in the gravid group, i.e., semigravid up to fully gravid. The annual fed/gravid ratios for PSC and PIT were similarly calculated, and the mean was used to determine the length of the gonotrophic cycle. This approach is based on the assumption that, disallowing for interim mortality, the fed and gravid categories should occur in approximately a 1:1 ratio for a 2-d gonotrophic cycle, whereas the ratio would approximate 1:2 for a 3-d cycle because of the existence of a semigravid state.

Entomological Inoculation Rate. The entomological inoculation rate was calculated as the product of the human biting rate and the sporozoite rate from LTC (equation 1):

$$\text{EIR} = 1.6 \times (\text{no. of sporozoite positive ELISAs from LTC}/\text{no. of mosquitoes tested}) \times (\text{no. of mosquitoes collected from LTC}/\text{no. of LTC}) \times 365 \quad [1]$$

An alternate calculation (equation 2) was done which assumes that all collected vectors are included in the sporozoite analysis, thereby eliminating the covariance between the sporozoite rate and the vector density (Drakeley et al. 2003). This approach allows calculation of confidence intervals around the estimate based on a Poisson distribution of sporozoite positive mosquitoes (equation 2):

$$\text{EIR} = 1.6 \times (\text{no. of sporozoite positive ELISAs from LTC})/(\text{no. of LTC}) \times 365 \quad [2]$$

Results

Climate. The year was divided into seasons based on temperature and rainfall trends observed over the 5-yr period preceding the study. Mean annual rainfall was 589 mm in the lower altitude villages, compared with 1,656 mm in the highlands. Rainfall over the entire altitude transect is typically bimodal, with a long rainy season in March–May and a shorter rainy season in October–November that is often unpredictable at lower altitudes. A cool, dry season occurs in June–August, after which temperatures rise, reaching a peak in January–February.

Species Composition. Over a 13-mo period, 820 LTC were performed in 30 houses selected from four villages. Ninety-seven percent of LTC were included in the final analysis with operator error and battery malfunction being the main reasons for exclusion. A total of 5,617 mosquitoes were caught, of these 3,294 (59%) were *An. gambiae* s.l., 316 (6%) were *An. funestus* s.l., 1,778 (32%) were culicines, and the remaining 229 (4%) were other anophelines (e.g., *Anopheles coustani* s.l. and *Anopheles rufipes rufipes* Gough).

In 20 houses selected from four villages, 563 PSC were performed over a 13-mo period. Ninety-eight percent of PSC were included in the final analysis with absence of house owner on catch date the main reason for exclusion. In total, 3,252 mosquitoes were caught in 549 PSC. Of these, 2,119 (65%) were *An. gambiae* s.l., 300 (9%) were *An. funestus* s.l., 825 (25%) were culicines, and eight (0.2%) were other anophelines.

During the same period, 339 PIT were performed in 12 pits located in four villages. Eighty-nine percent of PIT were included in the final analysis with intermittent flooding of pits the main reasons for exclusion. In total, 1,688 mosquitoes were caught in 303 PIT. Of these, 556 (39%) were *An. gambiae* s.l., 146 (9%) were *An. funestus* s.l., 975 (58%) were culicines, and 11 (1%) were other anophelines.

PCR Speciation. Eighty-five percent (374/442) *An. gambiae* s.l. specimens randomly selected from all seasons, altitudes, and trap types were identified successfully to species by using PCR. Of these, 373 (99.7%) were identified as *An. arabiensis* and a single individual (0.3%) as *An. quadriannulatus* s.l., whereas no *An. gambiae* s.s. were detected.

Vector Density. Repeated measures ANOVA was only performed on densities of *An. gambiae* s.l. from light trap catches because no vectors were collected by other methods at higher altitudes. For *An. gambiae* s.l. in light traps, there were significant main effects of time ($P = 0.0005$) and altitude ($P < 0.0001$) and a significant time*altitude interaction ($P < 0.0001$), indicating that changes in vector density at different altitudes did not occur simultaneously.

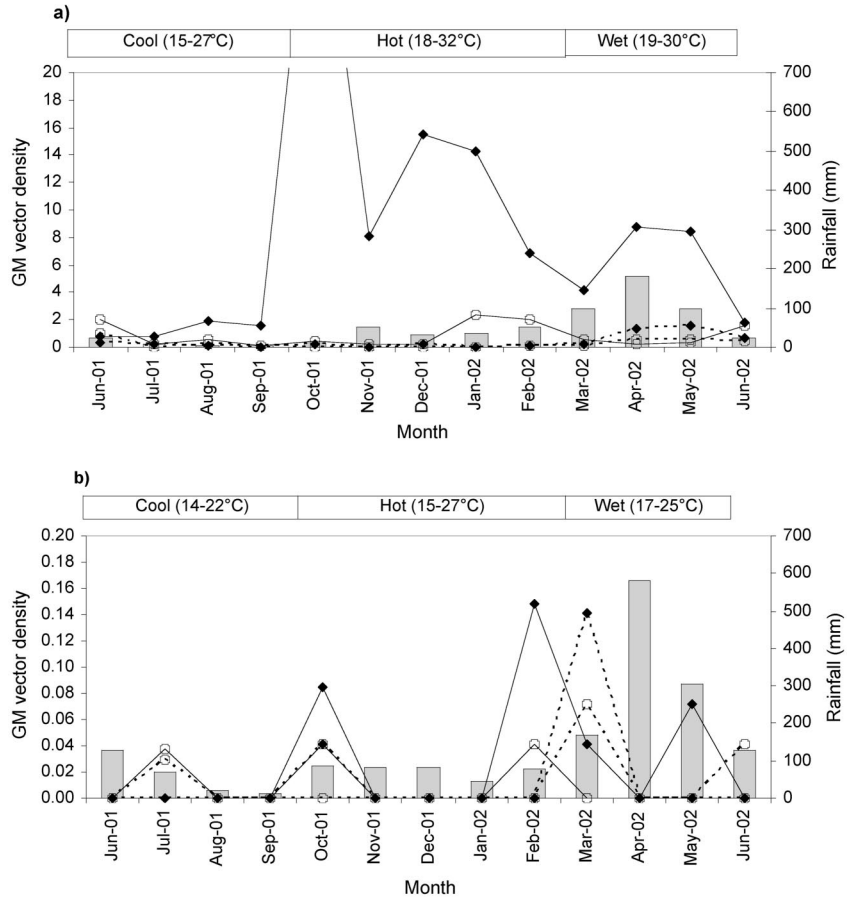


Fig. 2. Monthly GM densities of *An. arabiensis* (closed diamonds) and *An. funestus* s.l. (open circles) (GM number females per light trap per night), seasonal mean temperature ranges (shown in horizontal bars), and mean monthly rainfall (shaded columns) from June 2001 to June 2002 in the Hai District, Tanzania. (a) Lower and mid-altitude villages, 860 m, solid line for vector densities; 1,050 m, broken line for vector densities. (b) Upper altitude and highland villages, 1,350 m, solid line for vector densities; 1,565 m, broken line for vector densities.

House effect could not be tested as a class variable in the model because of insufficient degrees of freedom; however, clustering of catches was evident from frequency plots of house-wise monthly catch sums, indicating potential foci of transmission. These foci were related to proximity to breeding sites in the lower altitude villages, whereas in the highlands, this relationship was unclear. In LTC in Rundugai (860 m), 74% of all *An. gambiae* s.l. and 63% of all *An. funestus* s.l. were collected from two houses in adjacent subvillages, whereas the two spray catch houses neighboring these LTC houses accounted for 84% of all *An. gambiae* s.l. and 65% of all *An. funestus* s.l. caught resting indoors. These subvillages were located closest to the irrigated rice cultivation area. In Kware (1,050 m), 91% of *An. gambiae* s.l. and 81% of *An. funestus* s.l. from LTC were collected from one house, whereas the neighboring house accounted for 63% of *An. gambiae* s.l. and 84% of *An. funestus* s.l. caught by PSC. These houses were situated at the edge of the village, bordering a swamp area. In the upper altitude village, Sonu (1,350 m), vectors were collected from only two houses, with a single light trap house accounting for 91% of the total annual catch of *An. gambiae* s.l. In the highland village, Foo (1,565 m), a single light trap house collected 50% of *An. gambiae* s.l. and 75% of *An. funestus* s.l.

Monthly variation in nightly GM vector density from light traps is shown in Fig. 2. A regression of $\log_{10}$ annual GM vector density from light traps shows an exponential decrease with increasing altitude (Fig. 3), suggesting a 50% decrease in annual human biting rate for every 86-m increase in altitude.

Human Blood Index (HBI). Blood smears from 1,237 malaria vectors (1,095 *An. gambiae* s.l. and 126 *An. funestus* s.l.) were tested by precipitin test. The majority of fed vectors (n = 1,169; 95%) were caught in PSC and PIT in the village at 860 m. The remaining 18 (5%) were obtained from traps in the village at 1,050 m. Neither PSC nor PIT caught vectors in the two highland villages. Seasonal differences in HBI for indoor caught *An. gambiae* s.l. and *An. funestus* s.l. at

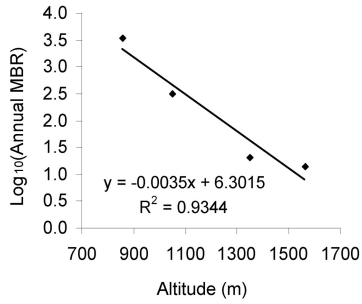


Fig. 3. Combined *An. gambiae* s.l. and *An. funestus* s.l. annual GM human-biting rate (MBR) estimated from light trap catches in four villages along an altitude transect in the Hai District, Tanzania, between June 2001 and June 2002. The annual MBR is presented on a log scale.

860 m were statistically significant ($\chi^2 = 22.9$, df = 2, $P < 0.001$ and $\chi^2 = 16.7$, df = 2, $P = 0.0002$, respectively) (Table 1). For *An. arabiensis*, the mean annual HBI was significantly higher in PSC than PIT at 860 m ($\chi^2 = 135$, df = 1, $P < 0.001$). Between villages, the mean HBI from PSC at 860 and 1,050 m was not significantly different ($\chi^2 = 0.36$, df = 1, $P = 0.57$). Annual HBI values were similar to those obtained by Ijumba et al. (2002) in the nearby Lower Moshi area of Tanzania, who used the ELISA technique of Burkot et al. (1981) for identification of bloodmeal source.

Sporozoite Rate. In total, 4,020 individual *An. arabiensis* and *An. funestus* s.l. mosquitoes were tested for *P. falciparum* circumsporozoite antigen by ELISA (Table 2). The sporozoite rate decreased from 0.36% (8/2198) at 860 m to 0% at 1,050 and 1,350 m. At 1,565 m, a sporozoite rate of 11.1% was found because of the presence of one positive individual out of a total of nine vectors collected. There were no differences between sporozoite rates from LTC, PSC, and PIT for *An. arabiensis* at 860 m ($\chi^2 = 4.7$, df = 2, $P = 0.09$) nor for *An. funestus* s.l. from PSC and PIT at 860 m ($\chi^2 = 0.3$, df = 1, $P = 0.57$) (Table 2). The main transmission season for *An. arabiensis* from light trap catches at 860 m was September to December (mean monthly sporozoite rate: s = 0.005, range 0.004–0.020), with a shorter transmission period in May (s = 0.004).

Entomological Inoculation Rate. A sharp decrease in EIR was observed along the altitude transect, from 47 ib/p/yr at the lowest of the altitude ranges to ≤2 ib/p/yr at higher altitudes; however, estimates at higher altitude ranges were associated with large variation because of the small sample sizes obtained. EIRs calculated from light trap catches using standard and alternate methods are presented in Table 3. Variation in monthly EIR in the lowest altitude village (860 m) showed two transmission seasons, corresponding with the hot and wet seasons. The peak monthly EIR in this village (19 infectious bites per person per month) occurred in early November.

Trap Efficiency and Fed/Gravid Ratio. On average, light traps caught 1.2 and 2.0 times as many *An. arabiensis* and 0.8 and 1.7 times as many *An. funestus* s.l. compared with spray catches in the lower and mid-altitude villages (860 and 1,050 m, respectively). Vectors were caught only by LTC at higher altitudes. There was significant seasonal variation in the relative efficiency of light traps and spray catches for catching *An. arabiensis* at 860 m ($\chi^2 = 49.9$, df = 2, $P < 0.001$) (Fig. 4). Fifty-six percent of the cool season catch of *An. arabiensis* was from PSC, whereas LTC supplied only 28%, suggesting increased endophily. In contrast, in the wet season, light traps were more effective than spray catches at collecting *An. arabiensis* (68 and 17% of seasonal catch, respectively), indicating reduced tendency to rest indoors. Pit traps consistently provided 15 to 16% of the total seasonal catch. The fed/gravid ratio for anopheline mosquitoes from indoor and outdoor catches can be used as an indicator of relative endophily versus exophily during the gonotrophic period. Decreased efficiency of spray catches relative to light traps in the hot and wet seasons coincides with increased proportions of gravid vectors in PIT relative to PSC. The seasonal changes in the fed/gravid ratio in indoor- and outdoor-resting populations (Fig. 4) therefore support the behavioral changes suggested by seasonal variation in trap efficiency.

Length of Gonotrophic Cycle. The fed/gravid ratio for *An. arabiensis* caught in PSC was 1:0.8 and 1:4.5 at 860 and 1,050 m, respectively, whereas the fed/gravid ratio in PIT was 1:1.9 at 860 m. The annual mean from

Table 1. Annual mean HBI and seasonal variation for vectors caught in spray catches and pit traps from two villages in the Hai District, Tanzania

Village	Season	*An. arabiensis*				*An. funestus* s.l.			
		Spray catch		Pit trap		Spray catch		Pit trap	
		n	HBI (%)	*n*	HBI (%)	*n*	HBI (%)	*n*	HBI (%)
Rundugai (860 m)	Mean annual HBI	905	73.9	144	25	86	67.4	34	0
	Cool	61	59.0	7	0	31	41.9	34	0
	Hot	711	72.9	135	27	36	75.0	0	–
	Wet	133	86.5	2	0	19	94.7	0	–
Kware (1,050 m)	Mean annual HBI	12	66.7	0	–	6	100	0	–
	Cool	7	57.1	0	–	6	100	0	–
	Hot	1	0	0	–	0	–	0	–
	Wet	4	100	0	–	0	–	0	–

n indicates the number of specimens tested.

Table 2. Annual *P. falciparum* sporozoite rate in vectors caught in LTC, PSC, and outdoor PIT in four villages situated along an altitude transect in Hai District, Tanzania

Village	Trap type	*An. arabiensis*			*An. funestus* s.l.		
		No. positive	No. examined	Positive (%)	No. positive	No. examined	Positive (%)
Rundugai (860 m)	LTC	8	2198	0.36	0	159	0
	PSC	4	621	0.64	1	133	0.75
	PIT	4	316	1.3	3	212	1.4
Kware (1,050 m)	LTC	0	203	0	0	84	0
	PSC	0	38	0	1	31	3.2
	PIT	0	1	0	0	1	0
Sonu (1,350 m)	LTC	0	14	0	0	2	0
	PSC	0	0	–	0	0	–
	PIT	0	0	–	0	0	–
Foo (1,565 m)	LTC	1	9	11.1	0	4	0
	PSC	0	0	–	0	0	–
	PIT	0	0	–	0	0	–

these two collection methods was 1:1.4 at 860 m, suggesting a 2- to 3-d gonotrophic cycle, and increased to 1:4.5 at 1,050 m, consistent with a 5- to 6-d cycle.

Discussion

The study confirms the predominant vector species in the study area was *An. arabiensis* with *An. funestus* s.l. of secondary importance. Overall, the density of *An. arabiensis* and *An. funestus* s.l. declined steeply with increasing altitude. Peak densities did not occur simultaneously across the altitude transect, likely because of differences in topography and large climatic variations of principle factors rainfall and temperature directly associated with elevation (Lindsay and Martens 1998, Minakawa et al. 2002). Low temperatures principally limit vector densities and hence transmission in the highlands (Lindsay and Martens 1998, Bødker 2000), whereas at lower altitudes the availability of surface water tended to limit vector population growth, and peak densities occurred during the rainy season and during periods of crop irrigation.

Seasonal succession of vector species was observed in the lower altitude villages. *An. arabiensis* density tended to increase at the onset of rainfall, whereas *An. funestus* s.l. growth experienced a lag of 3 to 4 wk. This difference can be explained by the differences in breeding habitat of the two vector species and has previously been observed where the two species overlap in distribution (Githeko et al. 1993, Charlwood et al. 1995, Minakawa et al. 2002). *An. funestus* s.l. may therefore assume a more important role in malaria transmission at the end of the long rainy season with its proportional density increase. *An. funestus* s.l. demonstrated variability in endophily and anthropophagy, suggesting that some of the specimens may not be *An. funestus* s.s. but rather another species of the *An. funestus* group, such as *Anopheles rivulorum* Leeson (Wilkes et al. 1996).

The presence of *An. quadriannulatus* s.l. at 1,050 m may indicate a focus of this nonvector member of the *An. gambiae* complex, which historically has been reported from Ethiopia, southern Africa, and the Zanzibar islands (White 1974, Gillies and Coetzee 1987); however, it is not known whether the specimen was truly the result of local breeding, and further investigation is recommended to confirm this finding.

Table 3. Annual EIR (infectious bites per person per year) from *An. arabiensis* and *An. funestus* s.l. in four villages situated along an altitude transect in Hai District, Tanzania, calculated using standard and alternate methods from light trap catches

Village (altitude)	EIR_{LTC}[a]	EIR_{LTC}[b] (95% CI)
Rundugai (860 m)	47	34 (14.7, 67.2)
Kware (1,050 m)	0	0 (0, 15.3)
Sonu (1,350 m)	0	0 (0, 8.4)
Foo (1,565 m)	2	2 (0.1, 12.4)

[a] Calculated by standard method (product of human biting rate and sporozoite rate).

[b] Calculated by alternative method; 95% CI based on Poisson distribution of sporozoite-positive mosquitoes.

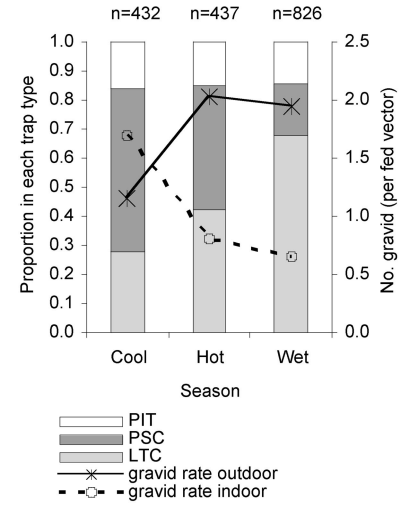


Fig. 4. Seasonal distribution of *An. arabiensis* catches by trap type (proportion of total seasonal catch in each trap type: PIT, PSC, and LTC) and corresponding seasonal changes in the gravid rate (number of gravid relative to fed vectors) collected indoors (PSC) and outdoors (PIT) in Rundugai village (860 m) in the Hai District, Tanzania.

Rapid population growth of *An. arabiensis* was observed at 860 m at the beginning of the hot season, reaching peak LTC density in October. This increase was similarly reflected in indoor- and outdoor-resting populations and was synchronized with rice (*Oryza* sp.) irrigation and increasing ambient temperatures at this altitude. Irrigated rice fields provide ideal breeding sites for members of the *An. gambiae* complex and are capable of generating vast numbers of mosquitoes (Ijumba and Lindsay 2001). Paradoxically, in areas of stable transmission, local communities near irrigated rice cultivation often experience lower levels of malaria infection in the human population than adjacent savannah communities (Boudin et al. 1992, Ijumba et al. 2002). However, in areas of unstable transmission where people have little or no immunity to malaria parasites, increased vector density after irrigation can lead to increased malaria in the human population (Ijumba and Lindsay 2001). The sharp rise in vector density at 860 m is followed by an increase in EIR; however, the degree of exposure of the local population may be reduced by use of personal protective measures. In this village, 66 and 17% of the population interviewed in a recent survey reported using bed-nets and mosquito coils/repellents, respectively (Swarthout 2002).

Resting behavior and host preference of *An. arabiensis* showed seasonal variation in line with previous studies (Molineaux et al. 1976, Petrarca et al. 1991, Bødker 2000). During the cool season the greater efficiency of PSC relative to LTC coincided with a decrease in the HBI of indoor-resting *An. arabiensis.* This relationship suggests an increased contribution of animal-fed mosquitoes to the spray catch because of entry of mosquitoes after feeding outdoors (Githeko et al. 1994), probably because of higher ambient temperatures inside houses, which may favor indoor resting during the cool season (Bødker 2000).

In contrast, in the hot season the efficiency of LTC relative to PSC increased and the HBI in PSC increased to 73%. During this period, vector densities were at their highest, and 27% of the *An. arabiensis* caught in PIT were found to have recently fed on human hosts, implying the exit of fed mosquitoes from houses (Sharp et al. 1990, Githeko et al. 1994). The shift in abundance of gravid mosquitoes from spray catches to pit traps supports postprandial exophily of *An. arabiensis* during this period.

In the wet season, there was a dramatic increase in the relative efficiency of LTC to PSC and a further increase in the HBI to 86%. The lower fed/gravid ratio in PSC during this period coincides with increased proportions of unfed and gravid *An. arabiensis* in PIT, compatible with a reduced exodus of females immediately after feeding and movement into outside resting sites in the middle of the gonotrophic cycle (Gillies 1954).

The gonotrophic cycle was longer at higher altitudes, based on the annual mean fed/gravid ratio from PSC and PIT, which takes into account the seasonal changes in resting behavior. Although these estimates assume equal accessibility of indoor and outdoor mosquito populations, they are nonetheless consistent with a previous estimate of a 3-d cycle in *An. arabiensis* at the lowest altitude range (Ijumba et al. 2002)

The lower sporozoite rates observed during the wet season are compatible with the trend of low female age during the population explosion of new emergence during the rains (Magesa et al. 1991), whereas at the end of the cool season the sporozoite rate reaches its peak when conditions are less favorable for breeding and the population is composed of older mosquitoes. The subsequent decrease in the sporozoite rate in the hot season is likely because of another large influx of young mosquitoes from increased vector breeding coinciding with crop irrigation.

Sporozoite-positive mosquitoes were more likely to be collected from spray catches in the cool season when *An. arabiensis* was more endophilic, whereas in the wet season infective mosquitoes were only collected from light traps and pit traps. This trend was observed at all altitudes. Calculation of inoculation rates from light traps used the conversion factor of 1.6 for human landing catches (Lines et al. 1991) to allow for comparison with other studies in northern Tanzania, although it should be noted that comparisons of these collection methods in different geographic locations with different vector populations have shown different relationships (Mbogo et al. 1993; Mathenge et al. 2005). Although sporozoite-positive mosquitoes were collected by indoor- and outdoor-resting catches, a reliable estimate of EIR could not be derived from either source because of exophily of *An. arabiensis* leading to underestimation of vector densities from PSC, and the inaccurate estimate of biting rates from PIT because of the unknown number of people available to be bitten by these vectors.

A study from the West Usambara Mountains documents a decease in EIR of 50% per 100 m from 91 ib/p/yr at 300 m to 0.03 ib/p/yr at 1,700 m, although values >1,000 m were extrapolated because of lack of sporozoite positives (Bødker 2000). Our estimate for 800 m is 10-fold higher than the corresponding altitude in the Usambaras, presumably because of the topography of the lower altitude area in this study, which is essentially flat and provides good breeding sites for vectors. Furthermore, these areas experienced excess rainfall during the study period, which likely contributed to an artificially high estimation of annual inoculation rates. The short rains, which are unpredictable from year to year, arrived in the study year coinciding with the hot season when sporozoite rates were highest, thereby causing a rapid increase in EIR. We also document a similar difference in highland EIR estimates between the Hai District and the Usambaras, although this is subject to large variation because of low mosquito densities. Most vectors collected at 1,565 m came from a single house at one time and although the reason for this variation is unknown, the presence of an infective vector in this catch is noteworthy.

Results of the current study contribute to the understanding of malaria transmission in this and other highland regions of Africa and suggest that measurement of transmission intensity at different altitudes will benefit from more than one collection method. Knowledge of vector behavior will allow more judicial

use of control interventions and may further influence the choice of control method used. Although year-round use of insecticide-treated nets would provide the best protection against vectors, the low mosquito densities in highland areas may not provide sufficient incentive for net use. Our data suggest that targeted indoor spraying during the cool season could be an effective means of reducing the residual *An. arabiensis* population, which is predominantly endophilic during this period. Behavioral changes also may influence selection pressure for insecticide resistance development, because increased endophily in the cool season would increase exposure to household insecticides. Importantly, seasonal endophily would be expected to facilitate malaria transmission because peak sporozoite rates occur at this time.

It may be considered that the low spatial intensity of mosquito sampling contributed to overestimation of inoculation rates in the present survey because only five houses were sampled by light trap in the lower and mid-altitude villages. Significant clustering of infective mosquitoes was observed in the lower altitude village, which influenced sporozoite rate estimation, because five of eight (62.5%) sporozoite-positive mosquitoes were collected from only two of 57 (3.5%) light trap catches. This clustering has been observed previously (Drakeley et al. 2003) and may be attributed to site fidelity of *An. arabiensis* whereby females return to the same houses to feed (McCall et al. 2001). Inclusion of more houses may therefore have resulted in a more representative estimation of the EIR; however, heterogeneities in malaria transmission often occur because of the nonrandom distribution of vectors (Burkot 1988), with a small proportion of households experiencing the majority of host–vector contact (Woolhouse et al. 1997), and additional sampling may have resulted in similarly variable estimates. In the highland areas, despite increased sampling intensity, insufficient vectors were collected to accurately estimate transmission, highlighting the need for more efficient collection methods at low vector densities. Widespread use of bed-nets and other personal protective measures may provide a further explanation for the lower than expected level of malaria morbidity. Further investigation of socioeconomic factors and spatial analysis of vector breeding sites along the altitude transect is therefore recommended.

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